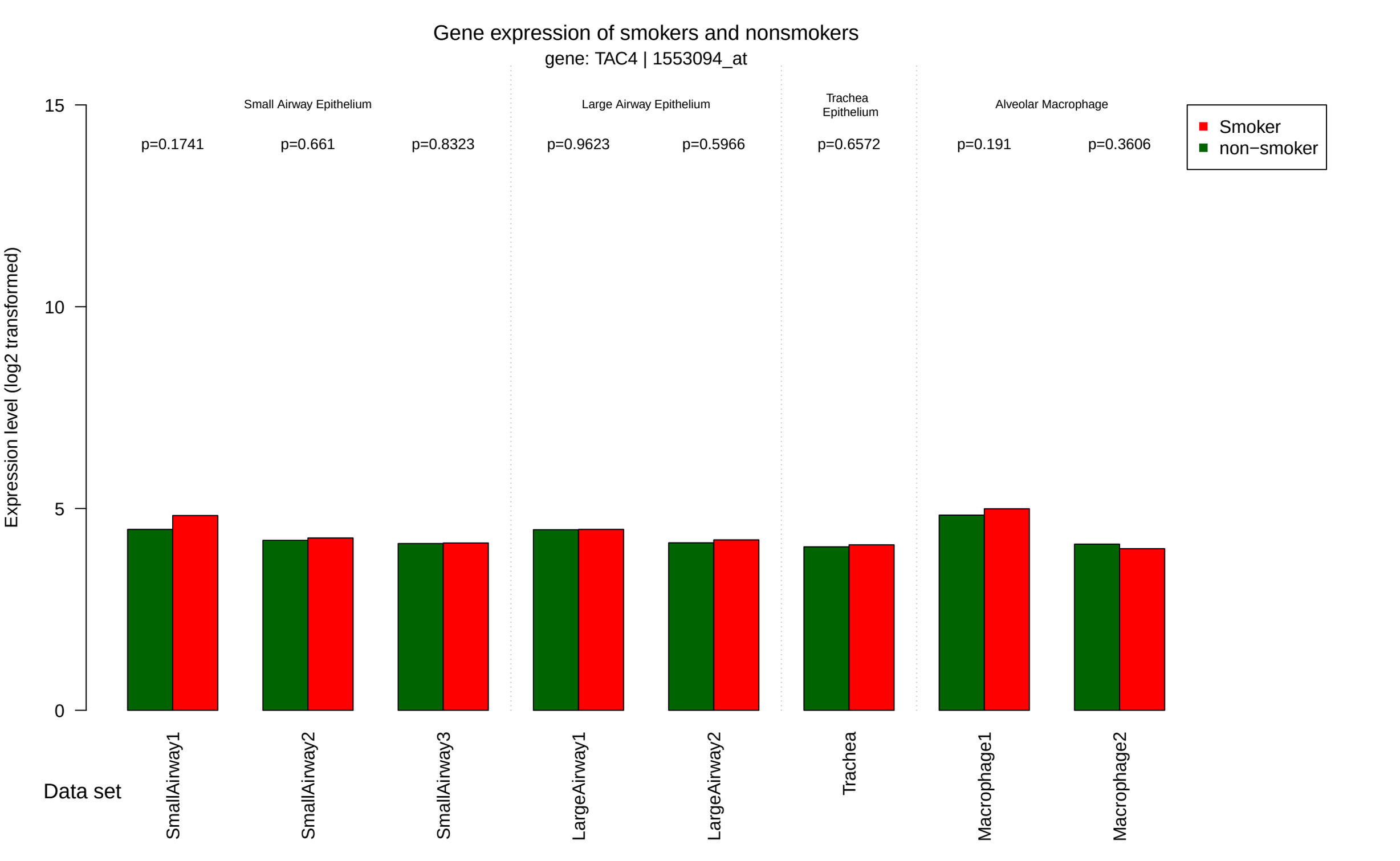
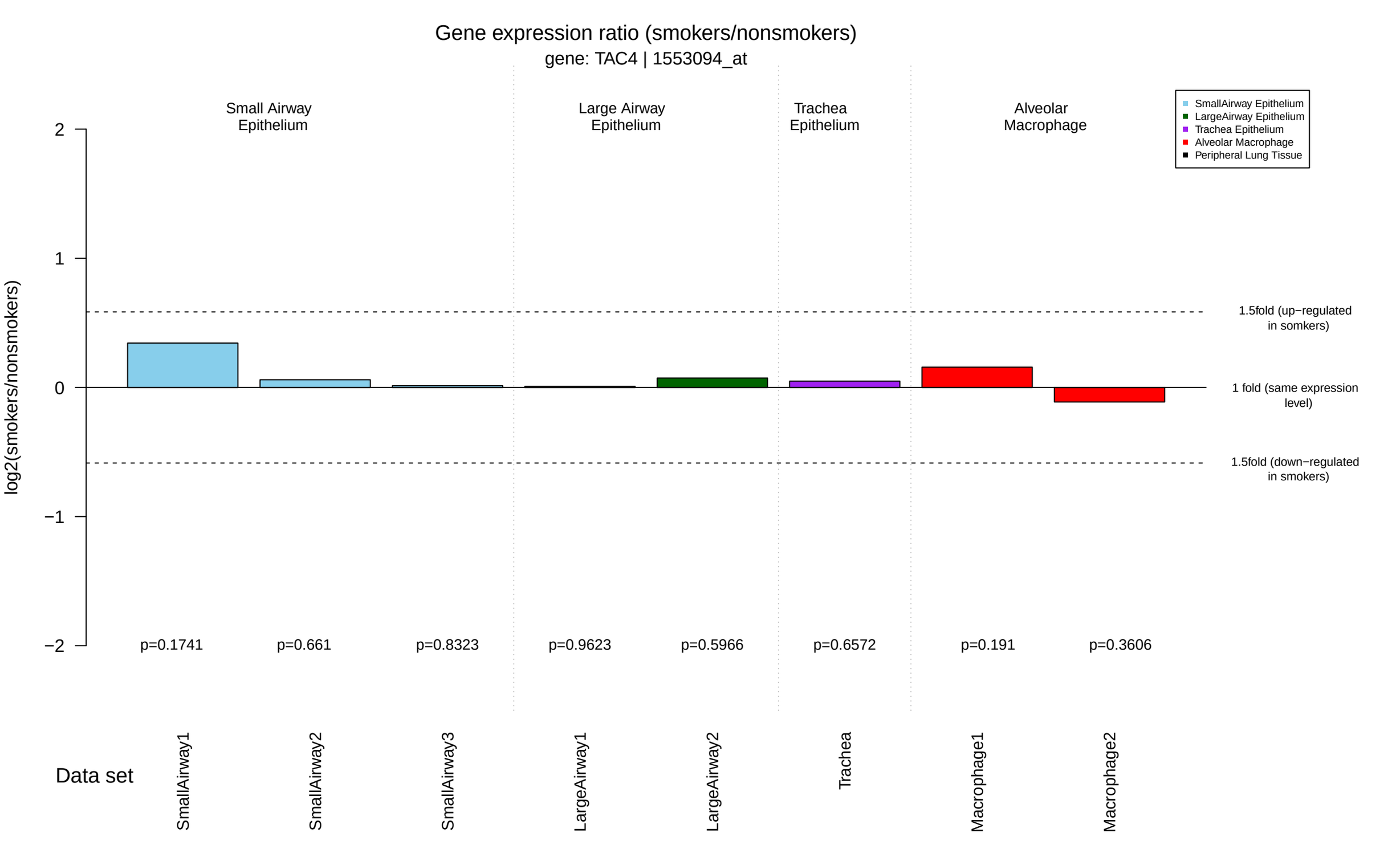
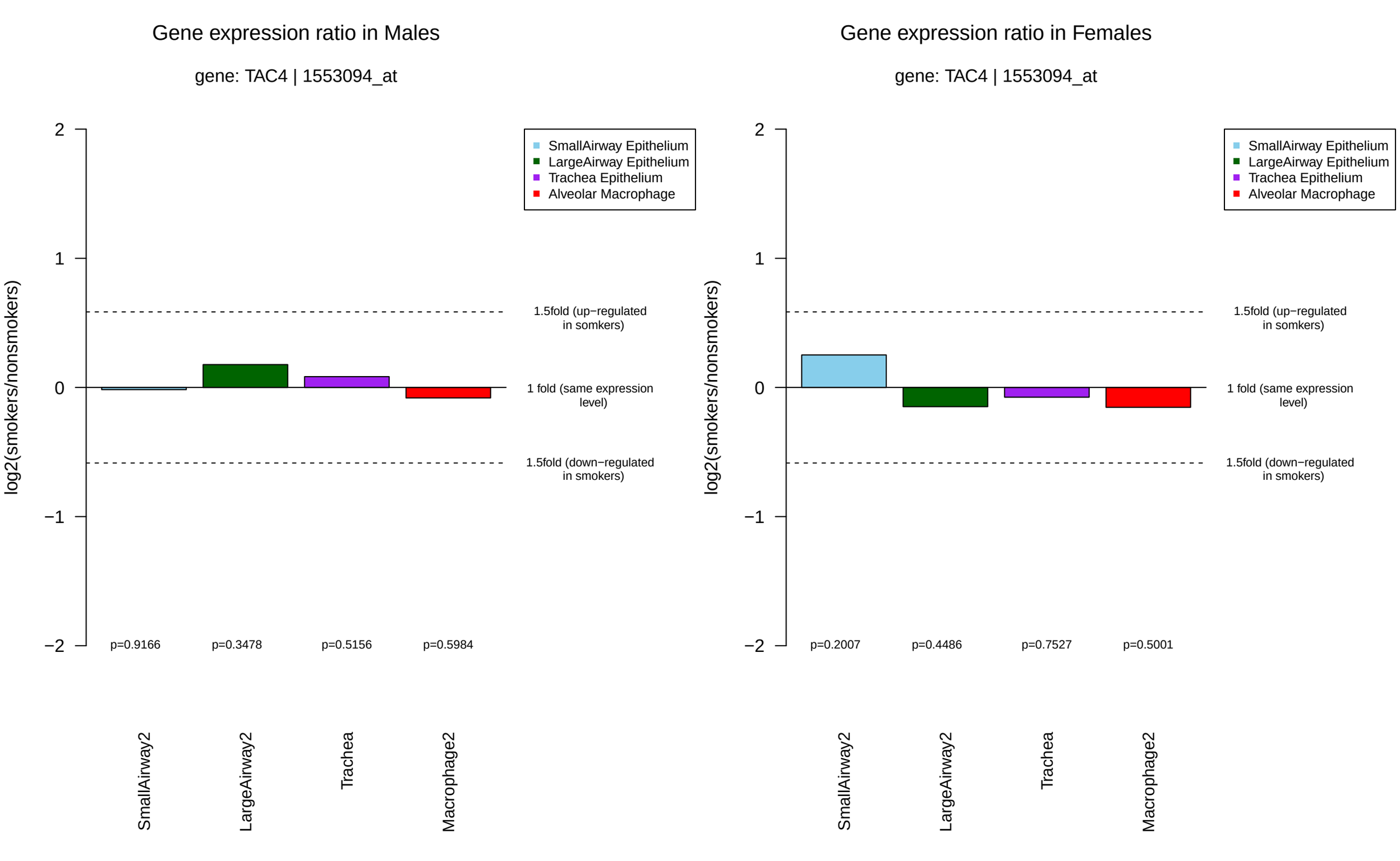
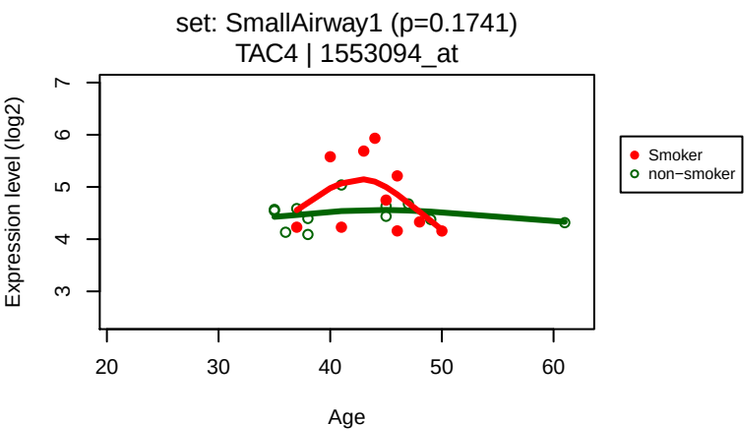
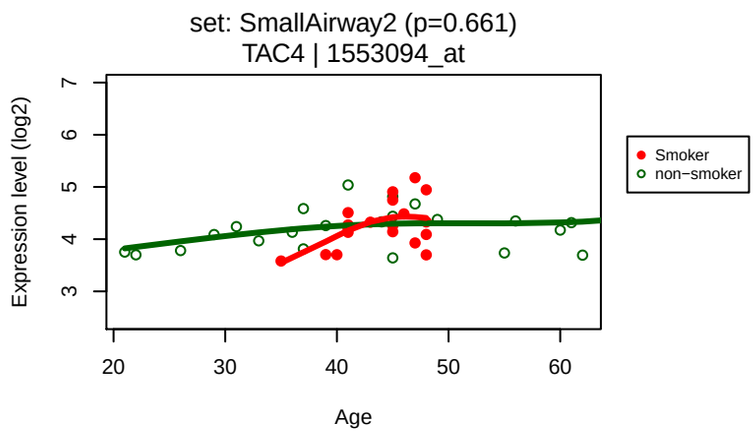
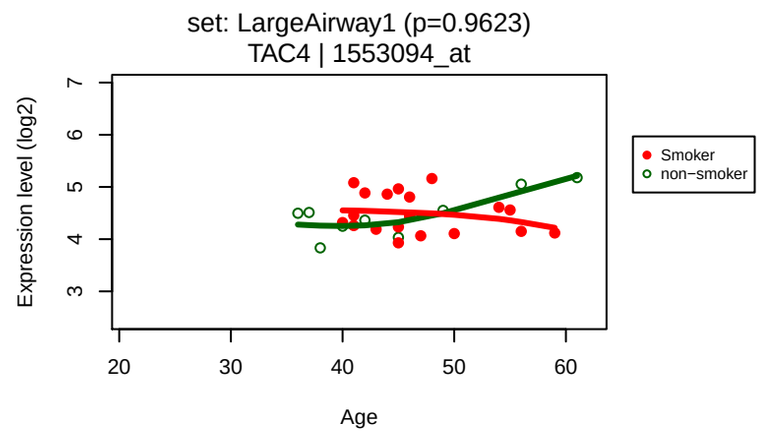

Gene expression of smokers and nonsmokers
gene: TAC4 | 1553094_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1741
p=0.661
p=0.8323
p=0.9623
p=0.5966
p=0.6572
p=0.191
p=0.3606
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

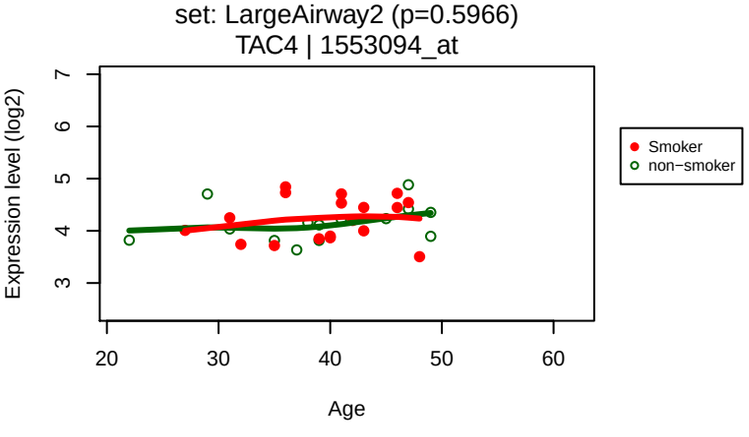

Gene expression ratio (smokers/nonsmokers)
gene: TAC4 | 1553094_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1741
p=0.661
p=0.8323
p=0.9623
p=0.5966
p=0.6572
p=0.191
p=0.3606
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

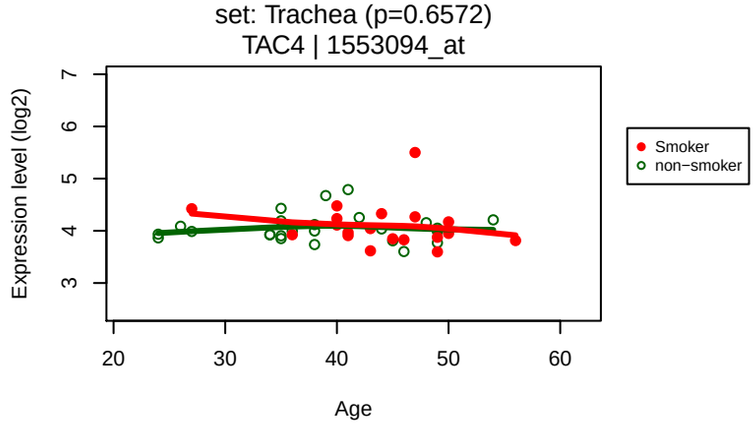

Gene expression ratio in Males
gene: TAC4 | 1553094_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9166
p=0.3478
p=0.5156
p=0.5984
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TAC4 | 1553094_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2007
p=0.4486
p=0.7527
p=0.5001
SmallAirway2
LargeAirway2
Trachea
Macrophage2

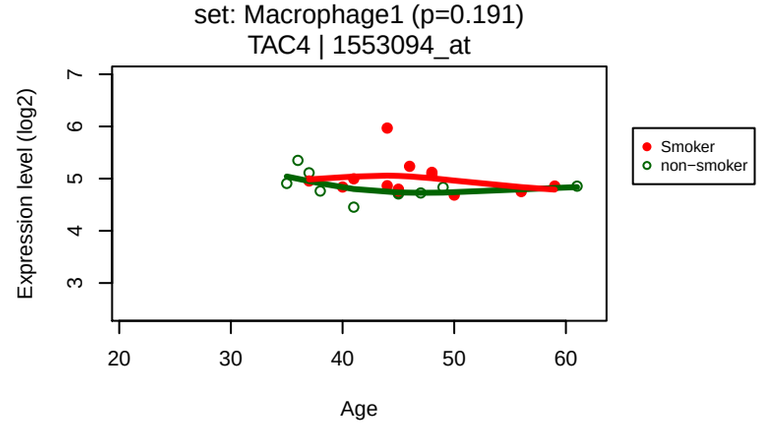

set: SmallAirway1 (p=0.1741)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker

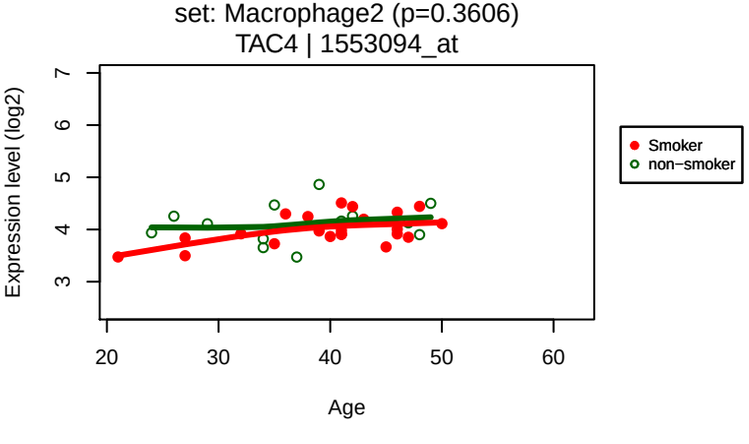

set: SmallAirway2 (p=0.661)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker

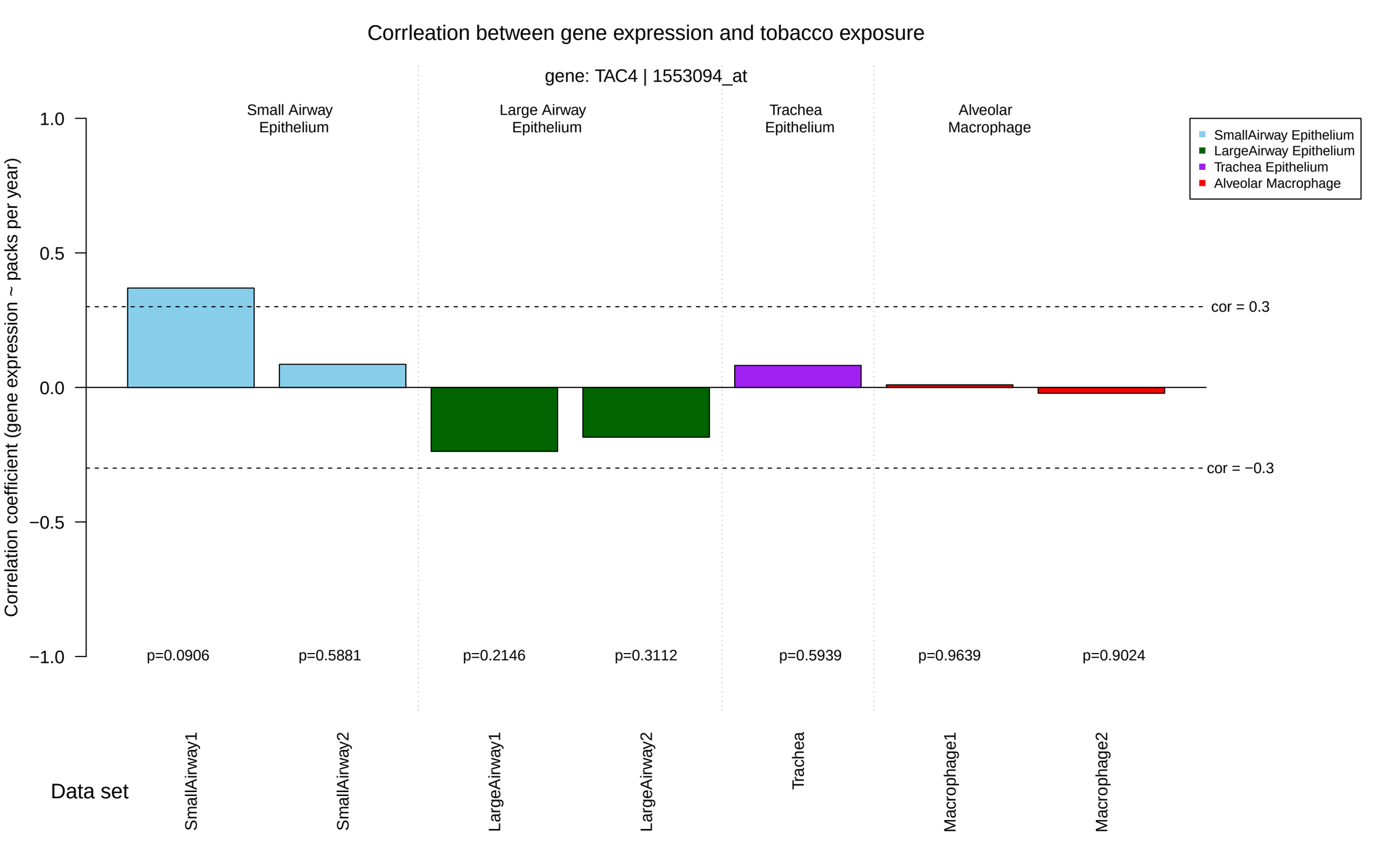

set: LargeAirway1 (p=0.9623)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5966)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6572)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.191)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3606)
TAC4 | 1553094_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TAC4 | 1553094_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0906
p=0.5881
p=0.2146
p=0.3112
p=0.5939
p=0.9639
p=0.9024
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2